

Retrieval-Stage Interactions in Cross-Modal Text-to-Image Retrieval Pipelines

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Abstract

Cross-modal retrieval systems enable searching content in one modality using queries from another. Recently, retrieving images using natural language queries – *text-to-image retrieval* – has received a lot of attention due to advances in deep learning vision models and interest in multi-modal processing in general. While several modeling approaches exist, ranging from simple keyword matching to joint embedding models and neural rerankers, there is limited understanding of how candidate generation, reranking, and fusion interact across retrieval pipelines.

In this paper, we present an analysis of retrieval-stage interactions in text-to-image retrieval pipelines, spanning a comprehensive set of categories drawn from recent advances in information retrieval: single index pipelines, multi-stage retrieve and rerank pipelines, and ensemble pipelines. We compare the effectiveness of these pipelines on the MS-COCO and Flickr30K benchmarks revealing several retrieval-stage phenomena governing reranking effectiveness, candidate-set expansion, and ensemble behavior.¹

CCS Concepts

• **Information systems** → **Retrieval effectiveness; Top-k retrieval in databases.**

Keywords

cross-modal retrieval; text-to-image retrieval; reranking; reciprocal rank fusion; retrieval pipelines

ACM Reference Format:

Sanskar S. Singh, Ziv Barretto, Bharati Khanijo, Aditya Sahu, and Maya Ramanath. 2026. Retrieval-Stage Interactions in Cross-Modal Text-to-Image Retrieval Pipelines. In *Proceedings of the 35th ACM International Conference on Information and Knowledge Management (CIKM '26)*, November 07–11, 2026, Rome, Italy. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3799682.3840060>

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¹Code: <https://github.com/data-iitd/mmir-cikm>



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ACM ISBN 979-8-4007-2539-5/2026/11

<https://doi.org/10.1145/3799682.3840060>

1 Introduction

Cross-modal retrieval systems enable searching content in one modality using queries from another. Among these, text-to-image retrieval has become an important workload due to advances in vision-language models and the increasing use of multi-modal search in retrieval systems, recommendation engines, and generative AI applications. Recent retrieval architectures span a broad spectrum, ranging from bi-encoder retrieval systems such as CLIP [8] and OpenCLIP [1] to expensive cross-encoder rerankers such as BLIP-2 [4] and Qwen3-VL [5]. Between these extremes lie late-interaction and instruction-tuned retrieval models that attempt to balance efficiency and retrieval effectiveness.

While many recent works focus on improving model architectures, comparatively less attention has been paid to understanding how the different stages of a retrieval pipeline interact with one another. In practice, modern retrieval systems are rarely single-stage systems. Instead, they are composed of multiple components. For example, a retrieve-and-rerank pipeline has an efficient first-stage retriever that retrieves candidates, which are then reranked by a reranker. Similarly, in ensemble pipelines, a fusion mechanism could aggregate ranked candidates from multiple retrievers. Therefore, the effectiveness of the overall system depends not only on the quality of the individual models, but also on the interaction between candidate generation and reranking.

The interaction becomes particularly important in multi-modal retrieval because retrieval architectures exhibit different retrieval behaviours. For example, larger candidate sets may improve retrieve-and-rerank pipelines by exposing more relevant candidates to the reranker, but may simultaneously degrade ensemble pipelines by introducing noisy, low-quality candidates into fusion methods such as reciprocal rank fusion (RRF) [2]. Further, gains obtained from reranking depend strongly on the quality of the first-stage retriever. That is, weak retrievers can benefit substantially from reranking, while strong retrievers may require equally strong rerankers in order to avoid accuracy degradation.

In this paper, we study these retrieval-stage interactions in text-to-image retrieval pipelines. Our goal is to characterise how candidate generation and reranking/fusion interact across different retrieval architectures. We evaluate representative pipelines spanning text-only retrieval, single-index multi-modal retrieval, retrieve-and-rerank pipelines, and ensemble pipelines using standard MS-COCO [7] and Flickr30K [15] baselines.

1.1 Contributions

Our analysis suggests that the behavior of multimodal retrieval pipelines is a result of candidate-set composition, reranker behavior, and retrieval diversity rather than standalone retriever accuracy. The following observations consistently support this across both benchmarks. We summarize them here.

1. First-stage retrieval recall saturates rapidly with increasing candidate depth, suggesting that many downstream improvements are constrained more by reranker behaviour than by candidate generation itself.
2. Reranking gains depend strongly on the quality of the first-stage retriever. Weak retrievers benefit substantially from reranking, while strong retrievers require strong rerankers to avoid degradation in accuracy.
3. Candidate-set size has opposite effects in retrieve-and-rerank and ensemble pipelines. Increasing the candidate set generally improves reranking pipelines, but can degrade ensemble pipelines due to noise.
4. Ensemble pipelines benefit from complementary query coverage between retrievers, although low-quality retrievers may still degrade overall retrieval effectiveness.

2 Setup

We consider three kinds of pipelines typically used in IR systems.

Single-index pipelines. Under this category of pipelines, we experiment with bi-encoder models to index images offline. We instantiate these three categories using representative bi-encoder, late-interaction, instruction-tuned, and cross-encoder retrieval models, including CLIP [8], FLAVA [10], OpenCLIP [1], UniIR [12], llama-nc [13].²

Retrieve-and-rerank pipelines. Standard retrieve-and-rerank pipelines choose a (relatively) fast first-stage retriever (BM25 [9] or a dense bi-encoder) to retrieve candidates, which are then reranked by a more computationally expensive reranker [6, 14]. We experimented with both bi-encoder and cross-encoder rerankers. Specifically, we use a subset of bi-encoders listed in the previous paragraph as rerankers and BLIP-2 [4] and Qwen3-VL [5] as cross-encoder rerankers. These span the cost spectrum: the bi-encoder rerankers are lightweight, *llama-nc* is a late-interaction (ColBERT-style) reranker, and BLIP-2 and Qwen3-VL are the expensive cross-encoders.

Ensemble pipelines. The idea of this set of pipelines is that different retrieval models or methods may have different strengths. In order to leverage those strengths, we retrieve top- k results from each method and combine the ranked results using reciprocal rank fusion (RRF) [2]. We study two-model bi-encoder ensembles to study fusion behaviour.

2.1 Datasets and Metrics

As mentioned earlier, our study is over MS-COCO and Flickr30K text-to-image retrieval benchmarks. Each image is associated with five captions. One caption is used as the retrieval query, thus giving

²Note that we use X_{Image} to denote the image embeddings of the model X and X_{Text} to indicate the text embeddings. In addition, UniIR provides a *joint* $\langle text, image \rangle$ embedding and is indicated by $UniIR_{Joint}$. Text embeddings are used only to index the captions of the images rather than the image itself.

one ground truth image per query. Retrieval effectiveness is evaluated using the standard Recall@ k ($R@k$, $k \in \{1, 5, 10\}$) metrics on the test split [3] of the MS-COCO [7] (5000 queries) and Flickr30K [15] (1000 queries) datasets.

2.2 Experimental Protocol

We study how retrieval-stage interactions affect overall pipeline behaviour and focus our analysis on the following: i) interaction between retrievers and rerankers, ii) effect of candidate-set size, iii) retrieval saturation behaviour with increasing candidate-set size, and iv) complementarity and overlap between retrievers in ensemble pipelines.

This study is confined to retrieval effectiveness; the latency, throughput and storage costs of these same pipelines are analysed in [11].

Unless otherwise specified, we vary the candidate-set size $k \in \{10, 30, 100\}$, with a default of $k = 100$ to study the effect of candidate expansion in both retrieve-and-rerank and ensemble pipelines.

2.3 Implementation and Reproducibility Details

All dense first-stage retrieval uses exact inner-product search (FAISS Flat) over 768-dimensional embeddings; reported accuracies reflect exact search rather than ANN approximation. Table 1 lists the model checkpoints used.

Model	Checkpoint / weights
CLIP	ViT-L/14@336px
OpenCLIP	ViT-L/14, laion2b_s32b_b82k
FLAVA	ViT-B/32
UniIR	ViT-L-14-quickgelu, clip_sf_large.pth
llama-nc	llama-nemoretriever-colembed-1b-v1
BLIP-2	blip2-itm-vit-g
Qwen3-VL	Qwen3-VL-Reranker-2B

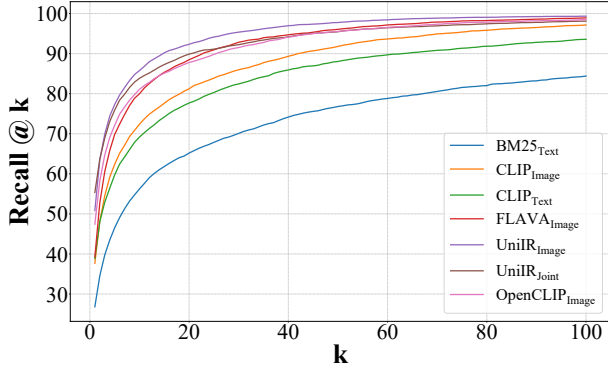
Table 1: Model checkpoints used across all pipelines.

Qwen3-VL-Reranker-2B reranks each of the top- k candidates independently. For each $\langle \text{query}, \text{image} \rangle$ pair, the model is prompted with the system instruction “*Judge whether the Document meets the requirements based on the Query and the Instruct provided. Note that the answer can only be ‘yes’ or ‘no’.*” and the task instruction “*Given a search query, retrieve relevant candidates that answer the query.*” The candidate is scored as the sigmoid-transformed difference between the language model head’s output logits for the tokens “yes” and “no”.

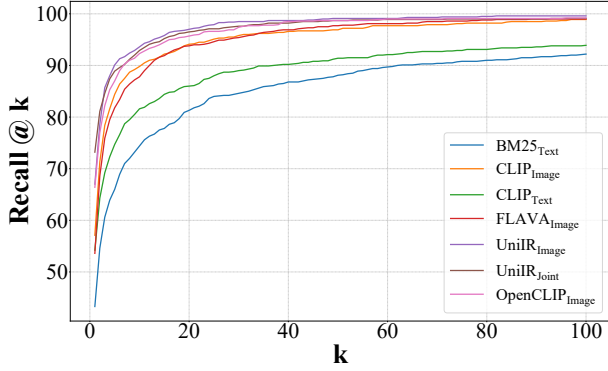
BLIP-2 scores each $\langle \text{query}, \text{image} \rangle$ pair with its image-text matching (ITM) head, and candidates are ranked by the resulting matching score.

Ensembles fuse two ranked lists via reciprocal rank fusion with the standard constant $K = 60$, applied over the top- k list from each retriever; each model contributes half of the candidates in the total candidate set.

The BM25 baseline is served by Apache Solr 9.8.1. For each dataset we build a single core from the stock `_default` configSet, which applies Lucene’s BM25 similarity with untuned parameters $k_1=1.2$ and $b=0.75$.



(a) MS-COCO



(b) Flickr30K

Figure 1: $Recall@k$ of first-stage retrievers as candidate depth k increases. These curves indicate the maximum achievable recall for downstream reranking pipelines.

All experiments ran on a single NVIDIA A100-PCIe-40GB GPU with 128 GB RAM; results reflect a single evaluation pass over the fixed test splits (retrieval is deterministic given the checkpoints).

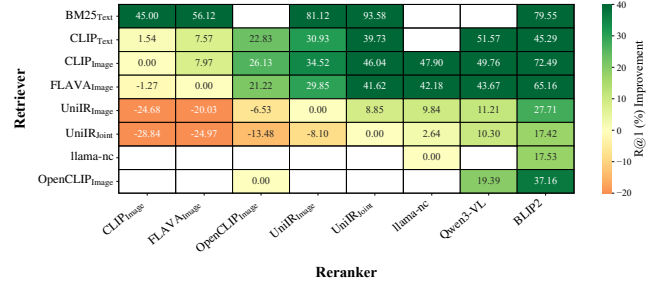
3 Retrieval-Stage Interactions

3.1 Candidate Recall Saturation

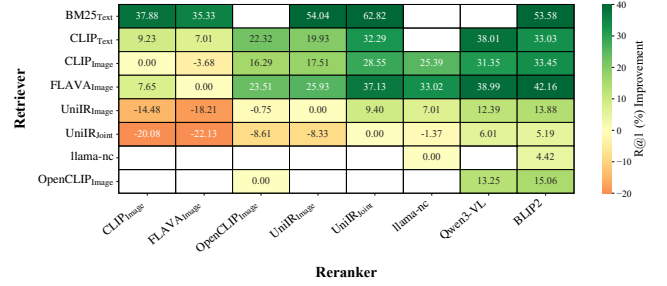
Figure 1 plots $Recall@k$ vs. k for each first-stage retriever pipeline in isolation (no reranking). This gives the “ceiling” that any downstream reranker or ensemble can aspire to since a reranker can only reorder the candidates.

Across both datasets, retrieval recall saturates rapidly as k increases. In particular, most retrievers recover more than 90% recall within relatively small candidate depths (often by $k \approx 30$ –50), after which additional candidates provide diminishing improvements. This indicates that many downstream retrieval differences are governed more by candidate ordering than by candidate availability.

This observation helps explain several downstream retrieval behaviours studied later in the paper. First, it explains why rerankers are effective even with relatively small candidate sets – because the relevant image is often already present within the top retrieved candidates. Second, it explains why differences between retrievers diminish after reranking – because, once candidate



(a) MS-COCO: Reference $R@1$ values for single-index retrieval - $BM25 = 0.268$, $CLIP_{Image} = 0.3766$, $FLAVA_{Image} = 0.3926$, $OpenCLIP_{Image} = 0.47$, $UniIR_{Joint} = 0.5534$, $llama-nc = 0.5546$



(b) Flickr30K: Reference $R@1$ values for single-index retrieval - $BM25 = 0.433$, $CLIP_{Image} = 0.571$, $FLAVA_{Image} = 0.536$, $OpenCLIP_{Image} = 0.664$, $UniIR_{Joint} = 0.7320$, $llama-nc = 0.714$

Figure 2: Percentage improvement in $R@1$ obtained by reranking a single-index baseline. Each row corresponds to a retriever (y-axis) and each column to a reranker (x-axis). A cell shows the relative change in $R@1$ when the retriever is reranked by the corresponding reranker.

recall becomes sufficiently high, reranker behaviour dominates end-to-end retrieval effectiveness. Hence, weak retrievers can still be effective in a retrieve-and-rerank pipeline, provided they retrieve a sufficient number of candidates that is passed on to a strong reranker.

More broadly, these results suggest that first-stage retrieval in these benchmarks appears constrained less by recall and more by candidate ordering and reranker quality.

3.2 Retriever-Reranker Interactions

We begin by analyzing how reranking effectiveness depends on the quality of the first-stage retriever. Figures 2a and 2b show the relative change in $R@1$ when different rerankers are applied on top of different retrievers. This behavior follows directly from the candidate-recall saturation observed in Figure 1. Once a retriever achieves sufficiently high candidate recall, downstream effectiveness becomes dominated primarily by candidate ordering rather than candidate availability.

Weak retrievers such as CLIP and FLAVA obtain large improvements from strong rerankers including BLIP-2, Qwen3-VL, and $UniIR_{Joint}$. In contrast, weaker rerankers provide limited gains and may even degrade already strong retrievers. For example, a

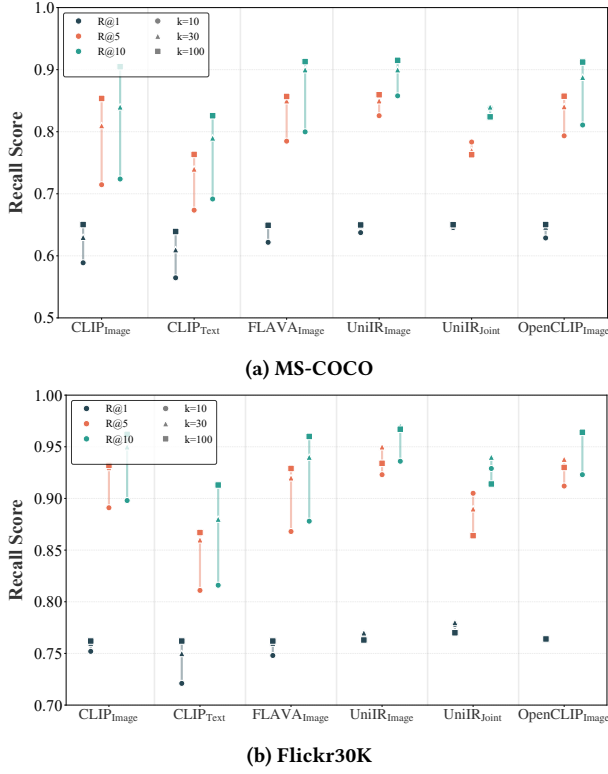


Figure 3: Retrieve-and-rerank pipeline recall when the first-stage candidate set size is increased from $k = 10$ to $k = 30$ to $k = 100$. Each group on the x-axis is a retriever whose output is passed to a BLIP-2 reranker; the three colours show $R@1$, $R@5$, and $R@10$. The general trend is that recall improves as the candidate set grows, i.e. rerankers benefit from seeing more candidates.

strong bi-encoder $\text{UniIR}_{\text{Joint}}$ benefits only from even stronger cross-encoder rerankers BLIP-2 and Qwen3-VL, and its performance degrades with weak rerankers like CLIP and FLAVA.

This clearly indicates that when the candidate set produced by the retriever already has highly relevant candidates near the top, a weak reranker may incorrectly reorder highly relevant candidates and degrade final retrieval quality. Conversely, weaker retrievers benefit substantially because strong rerankers compensate for poor candidate ordering.

Overall, these results indicate that reranking effectiveness is determined jointly by both retriever quality and reranker quality, rather than by reranker quality alone. Interestingly, as reranker strength increases, performance differences between first-stage retrievers diminish substantially, suggesting that strong rerankers can partially compensate for weaknesses in first-stage candidate ordering.

3.3 Effect of Candidate-Set Size

Next, we study how candidate-set size affects different retrieval architectures. Figures 3 and 4 report, respectively, how retrieve-and-rerank pipelines and ensemble pipelines respond to increasing the candidate set size from $k = 10$ to $k = 30$ to $k = 100$. For

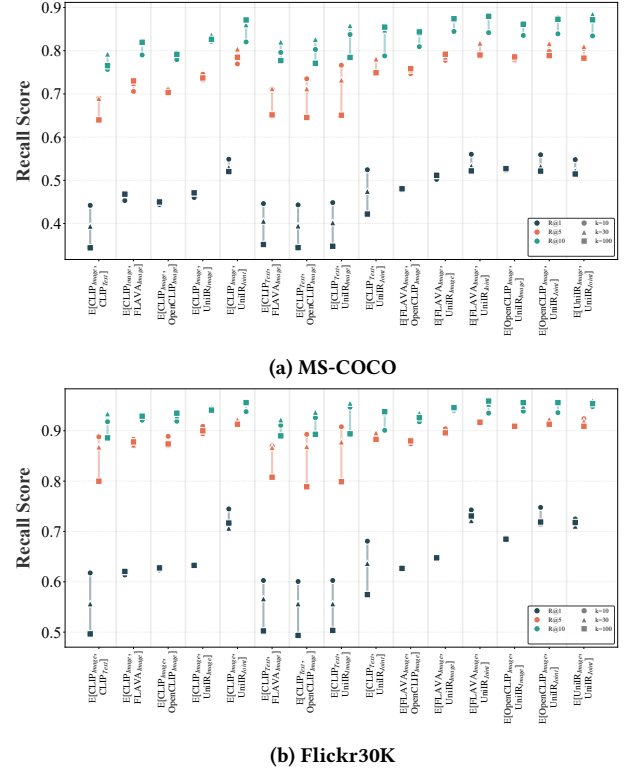
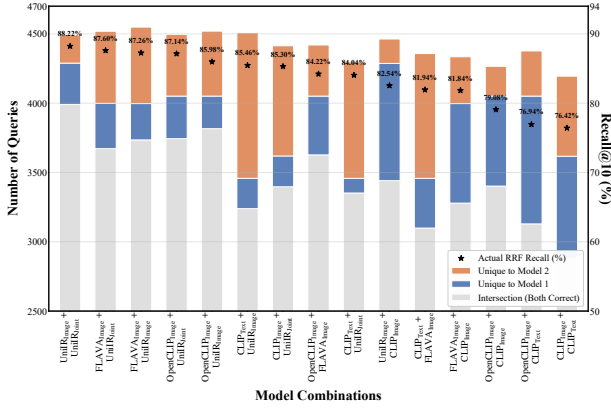


Figure 4: Ensemble pipeline recall when the candidate set size is increased from $k = 10$ to $k = 30$ to $k = 100$. Each group on the x-axis is an ensemble of two models ($E[\cdot, \cdot]$) fed into RRF; the three colours show $R@1$, $R@5$, and $R@10$. Increasing candidate depth improves ensemble Recall@10 but may simultaneously degrade Recall@1 due to noisy candidate expansion.

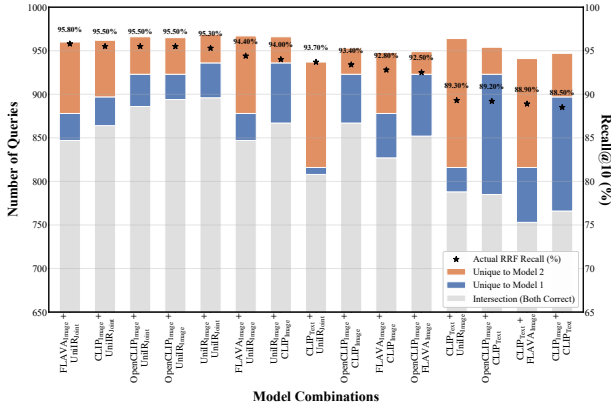
retrieve-and-rerank pipelines, recall generally *improves* as k grows: a larger candidate set gives the reranker more of a chance to recover the correct image. Importantly, rerankers can selectively reorder or suppress noisy candidates, allowing larger candidate pools to improve recall without necessarily degrading top-ranked results.

For ensemble pipelines, on increasing candidate depth $R@1$ often decreases while $R@10$ may improve. This suggests that RRF becomes increasingly sensitive to low-quality candidates contributed by weaker retrievers. As deeper candidate lists are fused, lower-ranked noisy candidates accumulate reciprocal-rank contributions across retrievers and may displace highly relevant candidates that were ranked near the top by the stronger retriever. For example, observe that $\text{CLIP}_{\text{Text}}$, a weak retriever ($R@1 = 0.3886$ for MS-COCO), when paired with a stronger retriever such as $\text{UniIR}_{\text{Joint}}$ ($R@1 = 0.5534$ for MS-COCO), consistently decreases the recall as it contributes more candidates, while the reranker substantially improves the recall of $\text{CLIP}_{\text{Text}}$ (as seen in Figure 3).

The different behaviours of these pipelines indicate that rerankers discriminate between potentially diverse candidates, and weak first-stage retrievers may still support highly effective retrieve-and-rerank systems, provided they achieve sufficiently high candidate recall for downstream rerankers. Further, fusion methods such as



(a) MS-COCO: Reference $R@10$ values for single-index retrieval - $CLIP_{Image} = 0.7238$, $FLAVA_{Image} = 0.7998$, $OpenCLIP_{Image} = 0.8108$, $UniR_{Joint} = 0.8384$.



(b) Flickr30K: Reference $R@10$ values for single-index retrieval - $CLIP_{Image} = 0.8980$, $FLAVA_{Image} = 0.8780$, $OpenCLIP_{Image} = 0.9230$, $UniR_{Joint} = 0.9290$.

Figure 5: Ensembles are referred to as Model 1 + Model 2 on the X-axis. The bars show query-level overlap between the two retrievers in each ensemble. Ensemble effectiveness depends strongly on complementary query coverage rather than standalone retriever accuracy alone. Actual recall ($R@10$) after RRF is indicated by the *.

RRF that simply aggregate rankings are vulnerable to low-quality retrievers.

Interestingly, while increasing candidate depth often degrades ensemble $R@1$, $R@10$ frequently improves. This suggests that larger candidate sets increase retrieval diversity and candidate coverage, but simultaneously introduce ranking instability near the top of the ranked list.

3.4 Complementarity in Ensemble Pipelines

We now analyse why ensemble pipelines improve retrieval effectiveness. Figure 5 presents an overlap analysis for different ensemble combinations. For each ensemble, we report the number of queries answered correctly only by the first retriever, only by the second retriever, and by both retrievers simultaneously.

The results show that ensemble gains arise primarily from complementary query coverage rather than uniformly improved ranking quality. Even retrievers with relatively similar $R@10$ values often retrieve different subsets of queries correctly. For example, in the MS-COCO dataset, the ensemble $FLAVA_{Image} + OpenCLIP_{Image}$ achieves a final $R@10 = 0.8422$ after RRF, improving over the standalone $R@10$ values of both (0.7998, 0.8108 respectively). Figure 5 shows that this improvement arises because each retriever answers a partially different subset of queries correctly. Consequently, ensembles benefit not merely from combining strong retrievers, but from combining retrievers with diverse retrieval behaviour.

4 Conclusions

Our study suggests that retrieval effectiveness depends critically on candidate-set composition and the interaction between candidate generation, reranking, and fusion. We showed that retrieve-and-rerank pipelines depend primarily on candidate recall and candidate ordering, while ensemble pipelines are influenced by retrieval diversity and are more sensitive to noisy candidate expansion. These observations highlight the importance of retrieval-stage interactions in determining end-to-end multimodal retrieval behaviour.

Acknowledgments

This work was funded by the R-Systems Centre of Excellence for Applied AI for Sustainable Systems.

GenAI Usage Disclosure

Generative AI tools were used to assist with writing and debugging experiment code, and editing the text of this paper.

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